Topological Full Configuration Interaction for Heteronuclear Diatomics: LiH as a Benchmark*

J. Loutey[1]

[1] *Independent Researcher, Kent, Washington*

(Dated: March 12, 2026)

We present a Topological LCAO architecture for heteronuclear full configuration interaction (FCI) within the GeoVac discrete graph framework. Two independent atom-centered hydrogenic lattices are concatenated into a bipartite molecular graph via Mulliken cross-nuclear attraction on the diagonal, Slater–Condon two-electron repulsion integrals with exact R^k radial integrals and Gaunt coefficients, and STO-overlap bridges on the off-diagonal; the resulting one- and two-electron Hamiltonian is diagonalized in the full Slater determinant space. For LiH at $n_{\max} = 3$ (28 spatial orbitals, 56 spin-orbitals, 367 290 determinants), the Boys–Bernardi counterpoise-corrected vertical binding energy at the experimental geometry ($R = 3.015$ bohr) is $D_e^{\mathrm{CP}} = 0.110$ Ha, within 20% of the experimental value $D_e = 0.092$ Ha. The balanced Hamiltonian produces a monotonically attractive potential energy surface with no equilibrium geometry, a limitation traced to the R-independence of the graph Laplacian's intra-atom kinetic energy: unlike continuous quantum mechanics, the discrete adjacency within each atomic subgraph does not respond to inter-nuclear compression. A Fock-weighted kinetic correction, whose *form* is derived from the energy-shell constraint $p_0^2 = Z^2/n^2$ of Paper 7 but whose overall scale λ is a single fitted parameter (retained as a labeled qualitative diagnostic, not a zero-parameter result), partially restores the equilibrium by concentrating orthogonalization repulsion on compact core orbitals, achieving $R_{\mathrm{eq}} \approx 3.8$ bohr and $D_e^{\mathrm{CP}} \approx 0.046$ Ha. A 29-version diagnostic arc (v0.9.8–v0.9.37) systematically tested and eliminated every competing hypothesis: basis set superposition error as a geometric driver, integral incompleteness, orbital exponent relaxation, SO(4) Sturmian bond sphere geometries, and Hamiltonian imbalance between cross-nuclear and electron repulsion orbital subspaces. In particular, Wigner D-matrix selection rules $D^2_{10,10}(\gamma) = 1$ and $D^2_{00,10}(\gamma) = 0$ are proven as structural results. The excitation-driven Direct CI solver achieves $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ scaling, preserving the $O(V)$ character of the underlying graph Hamiltonian.

I. INTRODUCTION

The GeoVac program posits that continuous quantum mechanics emerges from discrete graph topologies: a scale-invariant graph Laplacian defined on the unit three-sphere S^3 encodes the full single-particle Schrödinger equation without any continuous spatial integration [1–3]. Prior work established this correspondence rigorously for single-center atomic systems, recovering the hydrogenic spectrum as a combinatorial eigenvalue problem with $O(V)$ scaling. Extending this framework to multi-center molecules, however, presents a fundamental geometric challenge: two or more atomic lattices, each living on its own S^3 conformal manifold, must be bridged into a single molecular graph without resorting to the continuous $\mathbb{R}^3$ integration that the discrete premise is designed to replace.

In this work we introduce the *Topological LCAO* architecture, which solves the multi-center problem by concatenating atomic subgraphs into a bipartite molecular graph connected by discrete off-diagonal bridges. The resulting block Hamiltonian is fed into an $O(V)$ excitation-driven Full Configuration Interaction solver that operates entirely in Slater determinant space. As a headline result, the Boys–Bernardi counterpoise-corrected vertical binding energy of LiH at $n_{\max} = 3$ and $R = 3.015$ bohr is $D_e^{\mathrm{CP}} = 0.110$ Ha, within 20% of the experimental value of 0.092 Ha [8]. The balanced Hamiltonian reveals a fundamental limitation of the discrete LCAO framework: the absence of R-dependent kinetic repulsion, which is partially resolved by a Fock-weighted kinetic correction derived from Paper 7 [3].

LiH is the minimal four-electron heteronuclear diatomic and therefore the ideal first benchmark for the Topological LCAO framework. Its four electrons permit exact FCI treatment without truncation of the many-body expansion, while the $Z_A = 3$, $Z_B = 1$ asymmetry breaks the inversion symmetry exploited by homonuclear codes and exposes every heteronuclear coupling pathway. The experimental targets are well-established: $D_e = 0.092$ Ha and $R_{\mathrm{eq}} = 3.015$ bohr [8]. This paper formally extends the companion computational framework described in Paper 8 [4].

The paper is organized as follows. Section II develops the bipartite graph construction and the $O(V)$ Direct CI solver. Section III presents the LiH benchmark results, including the counterpoise-corrected potential energy surface and the Hamiltonian balance requirement. Section IV documents the 29-version diagnostic arc that systematically falsified every hypothesis for the observed behavior—integral incompleteness, orbital relaxation, and Sturmian geometries. Section V identifies the root cause as missing R-dependent kinetic repulsion

* GeoVac FCI Paper in the Geometric Vacuum series

and presents a Fock-weighted correction derived from Paper 7. Section VI summarizes and outlines next steps.

II. TOPOLOGICAL LCAO ARCHITECTURE

A. Subgraph Concatenation

The `MolecularLatticeIndex` instantiates two independent atomic subgraphs: a lithium lattice with nuclear charge $Z_A = 3$ and a hydrogen lattice with $Z_B = 1$. Each subgraph is a scale-invariant hydrogenic lattice defined on its own S^3 conformal manifold [3], retaining its atom-centered conformal structure and recovering, under the Fock projection, the hydrogenic spectrum $\varepsilon_n = -Z^2/(2n^2)$ (the bare graph Laplacian is positive-semidefinite; these negative energies are its continuum image, not graph-Laplacian eigenvalues). At $n_{\max} = 3$, the lithium subgraph contributes $\sum_{n=1}^{3} n^2 = 14$ spatial orbitals and the hydrogen subgraph contributes another 14, yielding a combined molecular basis of 28 spatial orbitals (56 spin-orbitals).

The concatenation is a direct sum at the graph level: the adjacency matrix of the molecular graph is block-diagonal in the two atomic subgraphs prior to bridging. No spatial overlap integral is computed between the subgraphs at this stage; the topological identity of each atomic lattice is preserved exactly.

B. Topological Bridging and the Hamiltonian

Bonding emerges from a rank-update concatenation that introduces off-diagonal coupling between the two atomic blocks. The one-electron molecular Hamiltonian takes the block form

$$\mathbf{H}_1 = \begin{pmatrix} \mathbf{H}_A & \mathbf{V}_{AB} \\ \mathbf{V}_{AB}^\dagger & \mathbf{H}_B \end{pmatrix}, \tag{1}$$

where $\mathbf{H}_A$ and $\mathbf{H}_B$ are the isolated atomic Hamiltonians and $\mathbf{V}_{AB}$ encodes the inter-atomic coupling.

The diagonal blocks carry the hydrogenic eigenvalues $\varepsilon_n = -Z^2/(2n^2)$ (the continuum image of the graph spectrum under the Fock projection, not graph-Laplacian eigenvalues) supplemented by the Mulliken cross-nuclear attraction: each orbital $\phi_i^{(A)}$ centered on atom A experiences the potential of nucleus B via the diagonal matrix element

$$\langle \phi_i^{(A)} | -Z_B/r_B | \phi_i^{(A)} \rangle, \tag{2}$$

evaluated analytically from the hydrogenic Fourier representation. Off-diagonal elements of $\mathbf{V}_{AB}$ are STO-overlap bridges weighted by conformal factors derived from the stereographic projection [5].

The two-electron repulsion V_{ee} is constructed via Slater–Condon rules with two classes of integrals. Same-atom Coulomb and exchange integrals use exact Slater R^k radial integrals with Gaunt angular coefficients, validated against the known analytical values $F^0(1s, 1s) = 5Z/8$, $F^0(1s, 2s) = 17Z/81$, and $F^0(2s, 2s) = 77Z/512$ [7]. Cross-atom Coulomb repulsion is evaluated via multipole expansion, with exchange restricted to s-orbital pairs where the overlap is non-negligible.

C. $O(V)$ Direct Configuration Interaction

Distributing four electrons across 56 spin-orbitals generates $\binom{56}{4} = 367\,290$ Slater determinants. A naïve pairwise construction of the CI Hamiltonian matrix would require $O(N_{\mathrm{SD}}^2)$ evaluations—over 10^{11} matrix element computations—rendering exact diagonalization impractical.

We instead adopt the excitation-driven Direct CI algorithm of Knowles and Handy [9], which constructs the Hamiltonian action $\sigma = \mathbf{H}|\Psi\rangle$ by traversing only the nonzero edges generated by valid single and double excitations from each determinant. The computational cost scales as $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$, where $N_{\mathrm{connected}}$ is the average number of determinants connected to each reference by at most a double excitation. For the LiH basis at $n_{\max} = 3$, this reduces wall time from hours to approximately 8 seconds when the Slater–Condon assembly loops are JIT-compiled via Numba [10] (approximately 75 seconds in pure Python).

The one-electron integrals h_{pq} and two-electron integrals $(pq|rs)$ are stored as dense arrays (56 spin-orbitals yield lookup tables of order 56^2 and 28^4, both fitting comfortably in memory), eliminating the $\sim 24\,\mu$s per-access overhead of `scipy.sparse` index validation that dominated earlier implementations.

D. Counterpoise Correction Framework

Basis set superposition error (BSSE) is an inherent artifact of any atom-centered expansion: at finite basis size, each atom can variationally borrow orbitals from its partner, artificially lowering the molecular energy below the true complete-basis-set limit. We implement the Boys–Bernardi counterpoise (CP) correction [6] natively within the Topological LCAO framework.

Ghost fragments are constructed by preserving the full bipartite topology of the molecular graph while zeroing the nuclear charge and electron population of the partner center. Specifically, a ghost-B calculation retains all 28 spatial orbitals but sets $Z_B = 0$, eliminating all bridges, cross-nuclear attraction, and cross-atom V_{ee} involving center B. The fragment energy E_A^{mol} is then computed in the identical topological basis space as the supermolecule, guaranteeing a consistent subtraction:

$$D_e^{\mathrm{CP}} = E_A^{\mathrm{mol}} + E_B^{\mathrm{mol}} - E_{AB}(R_{\mathrm{eq}}). \tag{3}$$

III. LIH BENCHMARK RESULTS

A. Computational Parameters

The molecular graph is assembled from two atomic subgraphs at $n_{\max} = 3$: a lithium lattice ($Z_A = 3$, 14 spatial orbitals) and a hydrogen lattice ($Z_B = 1$, 14 spatial orbitals), yielding a combined basis of 28 spatial orbitals and 56 spin-orbitals. The full four-electron Slater determinant space contains $\binom{56}{4} = 367\,290$ configurations. The exact ground-state eigenvalue is obtained from the excitation-driven Direct CI solver described in Sec. II C, with wall time of approximately 8 seconds on a single CPU core (Numba-accelerated assembly; approximately 75 seconds without JIT compilation).

B. Energetic Accuracy and Basis Set Superposition

Table I summarizes the key energetic results. The raw (uncorrected) binding energy significantly overestimates experiment, a clear signature of BSSE in the finite atom-centered basis. The Boys–Bernardi counterpoise correction quantifies the superposition error at $\Delta E_{\mathrm{BSSE}} = 0.115$ Ha, with the lithium fragment contributing 0.105 Ha and the hydrogen fragment 0.010 Ha—reflecting the far greater variational flexibility available to the smaller atom when it can access the partner's higher-Z orbitals.

After CP correction, the vertical binding energy at $R = 3.015$ bohr is $D_e^{\mathrm{CP}} = 0.110$ Ha, a 20% overestimate of the experimental value 0.092 Ha [8]. We emphasize that this is a *vertical* binding energy computed at fixed internuclear distance, not an adiabatic value at the theoretical equilibrium: the balanced Hamiltonian produces no equilibrium geometry (Sec. III C).

Quantity	Value
$n_{\max}$	3
Spatial orbitals	28
Spin-orbitals	56
Slater determinants	367 290
$E_{\mathrm{LiH}}(R{=}3.015)$ (Ha)	−8.118
$E_{\mathrm{Li}}^{\mathrm{mol}}$ (Ha)	−7.392
$E_{\mathrm{H}}^{\mathrm{mol}}$ (Ha)	−0.500
ΔE_{BSSE} (Ha)	−0.115
D_e^{raw} (Ha)	0.226
D_e^{CP} (Ha)	0.110
D_e^{expt} (Ha)	0.092
Error (%)	20

TABLE I. LiH energetics at $n_{\max} = 3$, $R = 3.015$ bohr (balanced `exact+True` configuration). The Boys–Bernardi counterpoise correction removes 0.115 Ha of basis set superposition error. The 20% overestimate of D_e is a vertical binding energy at fixed R; the balanced Hamiltonian produces no equilibrium geometry.

Composition note. The tabulated fragment energies $E_{\mathrm{Li}}^{\mathrm{mol}} = -7.392$ and $E_{\mathrm{H}}^{\mathrm{mol}} = -0.500$ Ha are the free-atom (own-basis) energies; substituted directly into Eq. (3) they yield the *raw* binding $D_e^{\mathrm{raw}} = 0.226$ Ha. The counterpoise correction evaluates the same fragments in the full dimer (ghost) basis—$E_{\mathrm{Li}}^{\mathrm{ghost}} = -7.497$, $E_{\mathrm{H}}^{\mathrm{ghost}} = -0.510$ Ha—so that Eq. (3) gives the counterpoise-corrected $D_e^{\mathrm{CP}} = D_e^{\mathrm{raw}} - \Delta E_{\mathrm{BSSE}} = 0.226 - 0.115 = 0.110$ Ha.

a. Hamiltonian balance requirement. The cross-nuclear attraction and electron repulsion must operate on matched orbital subspaces. Two balanced configurations—`exact` cross-nuclear with full cross-atom V_{ee} (`exact+True`), and `fourier` cross-nuclear with s-only V_{ee} (`fourier+s_only`)—produce molecular energies agreeing within 2 mHa across the entire PES, confirming variational consistency. An *imbalanced* configuration (`exact+s_only`) artificially overbinds by 0.55 Ha, as the p and d orbitals receive cross-nuclear attraction without the compensating electron repulsion screening. An earlier version of this work inadvertently used the imbalanced configuration; the results presented here are from the balanced `exact+True` Hamiltonian.

C. Absence of Equilibrium Geometry

The balanced configurations (`exact+True` and `fourier+s_only`) produce identical results within 2 mHa, confirming variational consistency. However, both yield monotonically attractive PES curves with no equilibrium geometry. The equilibrium reported in earlier versions arose from an imbalanced Hamiltonian configuration that has since been corrected.

The CP-corrected binding energy D_e^{CP} decreases smoothly from 0.464 Ha at $R = 1.5$ bohr to 0.042 Ha at $R = 5.0$ bohr, with the PES slope diminishing but never reversing sign. At no point does the repulsive kinetic wall balance the attractive cross-nuclear and exchange interactions: the system is variationally attracted to short bond lengths at all R.

This monotonic behavior is traced to a structural property of the Topological LCAO: the graph Laplacian kinetic energy within each atomic subgraph is *completely R-independent.* In continuous quantum mechanics, bringing two atoms together forces their electrons into higher-momentum orthogonal states, creating the kinetic repulsion that establishes the equilibrium bond length. In the discrete framework, each atom's adjacency matrix—and hence its contribution to the kinetic energy via the degree matrix D—is fixed at construction time and does not respond to the approach of a second center.

The only R-dependent kinetic contribution comes from the bridge edges connecting the two subgraphs. However, these bridge elements are approximately two orders of magnitude too weak: the maximum bridge matrix element at $R = 3.015$ bohr is 0.017 Ha, compared to the $\sim$0.5 Ha kinetic correction required by virial theorem analysis (the observed virial ratio $\eta = -V/(2T) \approx 30$ instead of the equilibrium value $\eta = 1$).

This absence of equilibrium geometry motivates both the systematic diagnostic arc in the following section and the kinetic repulsion analysis in Sec. V.

IV. SYSTEMATIC DIAGNOSTIC ARC

The initial implementation (v0.9.8) exhibited heavy overbinding ($D_e^{\mathrm{raw}} = 0.205$ Ha, more than twice the experimental value) and a 17% geometric contraction of the equilibrium bond length. Over the course of 29 controlled experiments spanning versions v0.9.8 through v0.9.36, every plausible hypothesis for these discrepancies was systematically tested. The experiments are grouped below by hypothesis class; the complete version history is tabulated in Appendix A.

A. Hypothesis 1: Basis Set Superposition Error (v0.9.8–v0.9.11)

The first hypothesis attributed both the energetic overestimation and the geometric contraction to BSSE: electrons borrowing orbitals from the partner center would artificially lower the molecular energy, inflating D_e and potentially distorting the PES minimum.

The Boys–Bernardi counterpoise correction (Sec. II D) quantified the superposition error at $\Delta E_{\mathrm{BSSE}} = 0.115$ Ha and corrected the vertical binding energy to $D_e^{\mathrm{CP}} = 0.110$ Ha (20% error). BSSE is the dominant source of energetic error in the raw binding energy but does not address the absence of an equilibrium geometry, which has a distinct root cause (Sec. V).

B. Hypothesis 2: Integral Incompleteness (v0.9.12–v0.9.13, v0.9.35)

The second hypothesis class attributed the geometric contraction to missing or approximate molecular integrals: either the Mulliken cross-nuclear attraction overestimates the short-range potential, or the restriction of cross-atom V_{ee} to s-orbital pairs neglects significant contributions from higher angular momentum channels.

Exact cross-nuclear attraction matrix elements, computed analytically from the hydrogenic Fourier representation and validated against the shell theorem, deviate by less than 0.6 mHa from the Mulliken diagonal approximation, confirming that the Mulliken scheme is essentially exact for this system.

Extending the cross-atom V_{ee} to all (n, l) pairs—392 Coulomb integrals at $n_{\max} = 3$ compared to 18 for the s-only restriction—contributes a correction of only +0.449 mHa to the total energy. This negligible shift reflects the fact that the LiH ground state is overwhelmingly s-orbital dominated: the $1s^2 2s^2$ configuration of lithium and the $1s$ of hydrogen leave higher angular momentum channels essentially unoccupied.

Integral incompleteness is conclusively falsified as a mechanism for the geometric contraction.

C. Hypothesis 3: Orbital Exponent Relaxation (v0.9.36)

The third hypothesis proposed that fixing atomic orbital exponents at their free-atom values forces artificial proximity: if electrons cannot polarize toward the bonding region by adjusting their radial extent, the system compensates by contracting R_{eq}.

A variational orbital exponent parameter ζ_B was introduced for the hydrogen center, allowing the effective nuclear charge $Z_{\mathrm{eff}} = \zeta_B Z_B$ to optimize the radial extent of all hydrogen-centered orbitals. The variational minimum occurs at $\zeta_B \approx 1.30$ at $R = 3.015$ bohr, relaxing toward unity at large R as expected.

The relaxation energy is approximately −0.27 Ha, but it is *uniform across the entire bonding region*: the energy lowering at $R = 2.5$ bohr and $R = 3.015$ bohr differs by only +0.007 Ha. Consequently, $\Delta R_{\mathrm{eq}} = 0.0$ bohr—the PES minimum does not shift. Moreover, the CP-corrected binding energy inflates by a factor of 4.7× to $D_e^{\mathrm{CP}} = 0.433$ Ha, demonstrating that the variational freedom is exploited for unphysical basis compression rather than physical polarization.

Orbital exponent relaxation is eliminated as a mechanism for geometric contraction.

D. Hypothesis 4: Bond Sphere and Sturmian Basis (v0.9.15–v0.9.34)

The fourth and most extensive hypothesis class proposed that the flat-space LCAO concatenation is fundamentally incompatible with the S^3 topology, and that a proper molecular treatment requires Sturmian basis functions defined on a single *bond sphere*—an S^3 manifold with two weighted poles—connected by SO(4) Wigner D-matrix elements.

This program was pursued across 20 versions (v0.9.15–v0.9.34), progressing from the initial SO(4) Wigner D-matrix implementation (44 tests, v0.9.15) through geometric-mean coupling (v0.9.16, unbound), Shibuya–Wulfman off-diagonal nuclear attraction (v0.9.17, unbound), hybrid Fourier–SW schemes (v0.9.18, 55% overbinding), atom-centered Sturmian variants (v0.9.20–v0.9.28, all unbound or massively overbinding), prolate spheroidal MO projections (v0.9.29–v0.9.30, unphysical), full MO Sturmian FCI (v0.9.31–v0.9.33, 12–25% error), and dual-p_0 extensions (v0.9.34, 25% error with p_0 values collapsing to a single geometric mean).

The definitive negative result is the structural theorem proven in companion Paper 8 [4] (Theorem 3): in any Sturmian basis sharing a single momentum parameter p_0, the one-electron Hamiltonian and overlap obey the two-term identity $H_{ij} = -\frac{1}{2}p_0^2 S_{ij} + (1-\beta_j)V_{ij}$ and share a single orthogonal $SO(4)$ Wigner-D congruence $D^{(n)}(\gamma(R))$ (so H is block-, not scalar-, proportional to S). The generalized eigenvalues are therefore independent of the bond angle γ, hence of R, yielding pure $1/R$ nuclear repulsion with no electronic binding. Recovering R-dependent binding energies requires solving the continuous two-center prolate spheroidal PDE for the spectrum $\beta_k(R)$, reintroducing exactly the continuous spatial geometry that the discrete graph framework is designed to eliminate. The Bond Sphere Sturmian alternative is mathematically falsified as a viable discrete framework.

a. Cross-n hybridization (v0.9.37). A further hypothesis attributed the geometric error to the D-matrix's block-diagonality in n: the Shibuya–Wulfman formula connects only same-n orbitals (Li $1s \leftrightarrow$ H $1s$, Li $2s \leftrightarrow$ H $2s$), giving zero off-diagonal coupling for the chemically relevant Li $2s \leftrightarrow$ H $1s$ pair. Overlap-based bridges using the Shibuya–Wulfman energy scale $-(Z_{\text{eff}}/p_0)\sin\gamma$ were implemented, producing Li $2s$–H $1s$ coupling of 0.052 Ha (40% of the same-n D-matrix coupling). However, the PES worsened: D_e increased from 7.1 to 10.7 eV because the cross-n coupling adds binding at short R where the overlap is largest, reinforcing the existing overbinding. Cross-shell hybridization is eliminated as a mechanism for the geometry error.

b. Level 4 locked-core (v0.9.37). The Level 4 molecule-frame hyperspherical solver (Paper 15) was extended to LiH by locking the Li $1s^2$ core and treating the two active electrons in hyperspherical coordinates with effective charges Z_{eff}. Four core models were tested (bare Z, integer screening, Clementi–Raimondi, and Core Hartree with penetration); all produce monotonically attractive PES with $R_{\text{eq}} \lesssim 2.5$ bohr and $D_e > 12$ eV. The failure has the same root cause: the core-to-H-nucleus attraction $V_{\text{cross}} = -2Z_B\langle 1s_Z^2|1/r_B|1s_Z^2\rangle$ behaves as $-2/R$ at large R but saturates exponentially at short R ($\exp(-6R)$ decay for Li $1s$), providing insufficient repulsion to establish a minimum at $R \approx 3$ bohr.

Having falsified BSSE as a geometric driver, eliminated integral incompleteness, ruled out orbital exponent relaxation, mathematically disproven the Sturmian alternative, excluded cross-n hybridization and locked-core hyperspherical approaches, the $O(V)$ Topological LCAO remains the only sound discrete framework for heteronuclear FCI. The absence of an equilibrium geometry in the balanced Hamiltonian is traced to the R-independence of the graph Laplacian kinetic energy, analyzed in the following section.

V. MISSING KINETIC REPULSION AND THE FOCK-WEIGHTED CORRECTION

A. The Kinetic Diagnostic

The virial theorem for a Coulombic system requires $\eta \equiv -V/(2T) = 1$ at the equilibrium geometry. Table II documents the virial ratio across the LiH PES: η ranges from 46 at $R = 1.5$ bohr to 22 at $R = 5.0$ bohr, indicating that the kinetic energy is $\sim 30\times$ too weak relative to the potential energy at all internuclear distances.

R (bohr)	T (Ha)	V (Ha)	$\eta = -V/(2T)$	E (Ha)
1.5	−0.092	−8.380	45.8	−8.471
2.5	−0.122	−8.055	32.9	−8.178
3.015	−0.133	−7.985	30.0	−8.118
4.0	−0.172	−7.908	23.0	−8.079
5.0	−0.181	−7.867	21.7	−8.049

TABLE II. Virial analysis of the balanced LiH PES. The virial ratio η should equal 1 at equilibrium but ranges from 22 to 46, indicating the kinetic energy is $\sim 30\times$ too weak.

Direct examination confirms that the intra-atom kinetic energy (the off-diagonal $\mathbf{H}_1$ within each atomic subgraph) is identically R-independent: the lithium and hydrogen graph Laplacians are constructed once from the isolated-atom topology and never modified as R changes. Bridge edges contribute a small R-dependent kinetic term through the degree matrix, but the maximum bridge element at $R = 3.015$ bohr is only 0.017 Ha, two orders of magnitude below the required repulsion.

This is a structural limitation of the Topological LCAO architecture: the discrete graph Laplacian has no mechanism to represent the orthogonalization kinetic energy that arises when atomic orbitals overlap in continuous quantum mechanics.

B. The Fock-Weighted Kinetic Correction

The energy-shell constraint of Paper 7—$p_0^2 = -2E = Z^2/n^2$ for the n-th shell—provides a natural scale for the "information density" of each orbital on the conformal S^3 manifold. We exploit this to construct a physically motivated kinetic correction.

For each orbital ϕ_a on atom A, the Fock-weighted overlap kinetic correction adds to the diagonal of $\mathbf{H}_1$:

$$\Delta h_{aa} = \lambda \sum_{b \in B} S_{ab}^2 \left(\frac{Z_A}{n_a}\right)^2 \left(\frac{Z_B}{n_b}\right)^2, \tag{4}$$

where $S_{ab} = \langle\phi_a|\phi_b\rangle$ is the STO overlap integral between s-orbitals on different centers, and λ is a single adjustable parameter. The Fock weights suppress diffuse orbitals (high n) and concentrate the kinetic penalty on compact core orbitals whose overlap decays rapidly with R.

The physical content of Eq. (4) is that the cost of overlapping two topological information structures scales with the product of their information densities $p_0^2 = Z^2/n^2$. Core–core pairs (Li 1s–H 1s) carry weight $9 \times 1 = 9$, while diffuse–diffuse pairs (Li 3s–H 3s) carry weight $1 \times 0.11 = 0.11$—an 81× discrimination.

The R-discrimination (ratio of correction at $R = 1.5$ to $R = 5.0$ bohr) improves from 3.0× with uniform weighting to 9.3× with Fock weighting. For the Li 1s orbital alone, the ratio improves from 19× to 74×.

C. Results of the Fock-Weighted Correction

Table III summarizes the λ-scan.

λ	Equilibrium?	R_{eq} (bohr)	D_e^{CP} (Ha)
0.00	No	—	—
0.02	No	—	0.091 at R=3.0
0.05	No	—	uptick at R=3.5
0.10	**Yes**	$\boldsymbol{\sim 3.8}$	**0.046**
0.20	Yes	~ 4.4	0.020
0.50	No (unbound)	—	—

TABLE III. Fock-weighted kinetic correction scan. At $\lambda = 0.1$, a genuine potential energy minimum emerges at $R \approx 3.8$ bohr with a repulsive wall below $R \approx 2.3$ bohr.

At $\lambda = 0.1$, the Fock-weighted correction creates a genuine potential energy minimum with a proper repulsive wall: the molecule is unbound below $R \approx 2.3$ bohr, maximally bound near $R \approx 3.8$ bohr ($D_e^{\mathrm{CP}} \approx 0.046$ Ha), and dissociates smoothly at large R. This is a qualitative success—the first equilibrium geometry produced by any balanced Hamiltonian configuration—but the single-parameter model cannot simultaneously reproduce both R_{eq} and D_e: at $\lambda = 0.02$, the CP-corrected binding energy matches experiment ($D_e^{\mathrm{CP}} = 0.091$ Ha at $R = 3.015$ bohr), but no equilibrium forms.

D. Remaining Limitations

The remaining discrepancy ($R_{\mathrm{eq}} = 3.8$ vs. experiment 3.015 bohr, $D_e = 0.046$ vs. 0.092 Ha) likely reflects the limitation of a purely diagonal correction. In standard quantum chemistry, kinetic repulsion includes both diagonal (orbital compression) and off-diagonal (orthogonalization) contributions from the full kinetic energy matrix $T_{AB} = \langle \phi_A | -\frac{1}{2}\nabla^2 | \phi_B \rangle$. The Fock-weighted correction captures only the diagonal component.

The BSSE of 0.115 Ha *strictly exceeds* the experimental dissociation energy of 0.092 Ha, confirming significant basis deficiency at $n_{\max} = 3$. However, since the balanced PES has no minimum, BSSE is not the mechanism for geometric contraction—it is a separate issue of basis completeness that CP correction addresses for vertical binding energies.

VI. CONCLUSION

We have presented the first heteronuclear full configuration interaction calculation on a discrete graph Hamiltonian. The Topological LCAO architecture concatenates atom-centered hydrogenic lattices via Mulliken cross-nuclear attraction, Slater–Condon two-electron integrals, and STO-overlap bridges, and diagonalizes the resulting Hamiltonian in the full 367 290-determinant Slater space using excitation-driven Direct CI.

For LiH at $n_{\max} = 3$, the Boys–Bernardi counterpoise-corrected vertical binding energy at $R = 3.015$ bohr is $D_e^{\mathrm{CP}} = 0.110$ Ha, within 20% of the experimental value of 0.092 Ha. The balanced Hamiltonian produces a monotonically attractive PES with no equilibrium geometry—a limitation traced to the R-independence of the graph Laplacian's intra-atom kinetic energy, not to basis truncation.

A 29-version diagnostic arc (v0.9.8–v0.9.37) systematically falsified every competing hypothesis: basis set superposition error as a geometric driver, integral incompleteness, orbital exponent relaxation, and the SO(4) Sturmian Bond Sphere alternative. The missing kinetic repulsion was identified as a structural limitation of the LCAO concatenation, and a Fock-weighted kinetic correction derived from Paper 7's energy-shell constraint $p_0^2 = Z^2/n^2$ partially restores the equilibrium ($R_{\mathrm{eq}} \approx 3.8$ bohr, $D_e^{\mathrm{CP}} \approx 0.046$ Ha at $\lambda = 0.1$).

The $O(V)$ scaling of the underlying graph Hamiltonian is preserved throughout, with the Direct CI solver achieving $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ assembly cost. The natural next steps are:

1. Extending the Fock-weighted correction to include off-diagonal kinetic matrix elements between atomic centers, which would provide the missing intermediate-R binding;
2. Investigating whether the Bond Sphere geometry of Paper 8 naturally incorporates the required R-dependent kinetic coupling through the SO(4) Wigner D-matrix;
3. Increasing $n_{\max}$ to assess basis convergence of the vertical binding energy.
4. **Prolate spheroidal lattice:** The fundamental limitation identified above—that the LCAO graph Laplacian provides no R-dependent kinetic repulsion—arises because the molecular graph is constructed as a concatenation of atomic S^3 subgraphs whose adjacency matrices are fixed at construction time. This limitation is resolved by recognizing that the natural geometry for two-center systems is the prolate spheroid, not the concatenated S^3. In prolate spheroidal coordinates (ξ, η, φ), the two-center Coulomb problem separates exactly, and discretizing the Laplace-Beltrami operator on this manifold yields a molecular lattice with R-dependence built into the metric. For H_2^+, this

approach achieves $R_{\rm eq} = 2.001$ bohr (0.21% error) at the finite-difference level, while the spectral Laguerre radial solver reaches 0.0002% energy error, both with zero fitted parameters [11]. The LCAO framework documented in this paper remains valuable for understanding the anatomy of molecular binding in a discrete framework, and the diagnostic arc (Table IV) provides a systematic record of approaches that fail and why.

The discovery that the Fock projection's energy-shell scale provides the correct weighting for kinetic repulsion reinforces the physical content of the conformal S^3 framework: the momentum parameter $p_0^2 = Z^2/n^2$ is not merely a mathematical artifact of the stereographic projection but encodes the "information density" of each quantum state on the three-sphere.

Appendix A: Version History

Table IV documents the complete diagnostic arc from v0.9.11 (the counterpoise-corrected baseline) through v0.9.38, including architecture, results, and root cause classification for each version.

[1] J. Loutey, "Angular Momentum as Information Geometry: Isotropic Packing and the Universal Angular Cross-Section," GeoVac Paper 0 (2026).

[2] J. Loutey, "The Geometric Atom: Atomic Structure as a Packing Problem," GeoVac Paper 1 (2026).

[3] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," GeoVac Paper 7 (2026).

[4] J. Loutey, "Bond Sphere: Diatomic Molecules as S^3 and their Natural Sturmian Basis," GeoVac Papers 8–9 (2026).

[5] V. Fock, "Zur Theorie des Wasserstoffatoms," Z. Phys. **98**, 145 (1935).

[6] S. F. Boys and F. Bernardi, "The calculation of small molecular interactions by the differences of separate total energies. Some procedures with reduced errors," Mol. Phys. **19**, 553 (1970).

[7] H. Shull and P.-O. Löwdin, "Superposition of Configurations and Natural Spin Orbitals. Applications to the He Problem," J. Chem. Phys. **30**, 617 (1959).

[8] K. P. Huber and G. Herzberg, *Molecular Spectra and Molecular Structure IV: Constants of Diatomic Molecules* (Van Nostrand Reinhold, New York, 1979).

[9] P. J. Knowles and N. C. Handy, "A new determinant-based full configuration interaction method," Chem. Phys. Lett. **111**, 315 (1984).

[10] S. K. Lam, A. Pitrou, and S. Seibert, "Numba: A LLVM-based Python JIT compiler," in *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC* (ACM, 2015), pp. 1–6.

[11] J. Loutey, "The Molecular Fock Projection: Diatomic Molecules as Prolate Spheroidal Lattices," GeoVac Paper 11 (2026).

Version	Architecture	D_e^{CP} (Ha)	R_{eq} (bohr)	Status	Root Cause
v0.9.11	LCAO baseline (balanced `exact+True`)	0.110	No eq.	Baseline (20%)	CP-corrected vertical D_e
v0.9.12	Löwdin orthogonalization	—	—	Fail	Breaks variational principle
v0.9.13	Mulliken exchange K	0.110	$\sim$2.5	19% err	K overcorrects D_e
v0.9.15	SO(4) Wigner D-matrix	—	—	Infrastructure	44/44 tests, no production
v0.9.16	D-matrix cross-nuclear (geom. mean)	—	—	Unbound	Coupling too weak
v0.9.17	Shibuya–Wulfman off-diagonal	—	—	Unbound	SW coupling too weak
v0.9.18	Hybrid (Fourier diag + SW off-diag)	0.143	<2.0	55% err	Dissociation limit wrong
v0.9.20	Sturmian diagonal	—	—	Unbound	1s diagonal positive
v0.9.21	p_0-consistent D-matrix + self-loop	—	—	Unbound	Compactness overbinding
v0.9.22	Atom-dependent p_0	3.777	$\sim$2.0	3.8× overbind	Scales do not decouple
v0.9.23	Shell-radius attenuation	—	—	Negligible	Correction $<$ 1 mHa
v0.9.24	Energy decomposition diagnostic	—	—	Tool	Identifies Fourier diagonal
v0.9.25	Ohno–Klopman cross-atom V_{ee}	—	—	Negligible	Cross-atom J irrelevant
v0.9.26	Angular-weighted V_{ee}	—	—	Negligible	$\Delta E \sim 0.4$ mHa
v0.9.27	SO(4) CG cross-center V_{ee}	—	—	Unbound	$J_{\mathrm{cross}} > J_{\mathrm{same}}$
v0.9.28	Bond sphere node weights	—	—	Unbound	β_H 46% short
v0.9.29	Prolate spheroidal MO betas	—	—	Unbound	LiH mapping ad hoc
v0.9.30	MO-to-atom projection	—	—	Unphysical	$E = -42.5$ Ha (5.3×)
v0.9.31	MO Sturmian FCI (full)	—	—	Unbound/overbinding	Shared $SO(4)$ congruence, eigenvalues γ-ind
v0.9.32	MO Sturmian FCI (variant)	—	—	Unbound/overbinding	Shared $SO(4)$ congruence, eigenvalues γ-ind
v0.9.33	MO Sturmian FCI (variant)	—	—	Unbound/overbinding	Shared $SO(4)$ congruence, eigenvalues γ-ind
v0.9.34	Dual-p_0 extension	—	—	Unbound/overbinding	Shared $SO(4)$ congruence, eigenvalues γ-ind
v0.9.35	All-l cross-atom V_{ee}	0.110	No eq.	$\Delta E = 0.4$ mHa	Negligible correction
v0.9.36	Orbital exponent relaxation	0.433	No eq.	4.7× overbind	Uniform PES shift
v0.9.37	Hamiltonian balance diagnostic	0.110	No eq.	Corrected	Imbalance identified
v0.9.38	Numba-accelerated assembly	0.110	No eq.	738× speedup	Python loop overhead eliminated

TABLE IV. Diagnostic arc version history (v0.9.11–v0.9.38). The LCAO baseline (v0.9.11, balanced `exact+True`) achieves correct-order-of-magnitude binding ($D_e^{\mathrm{CP}} = 0.110$ Ha, 20% error) but lacks equilibrium geometry due to missing kinetic repulsion.